

AUGUMENTED AND VIRTUAL REALITY IN CONSTRUCTION

Seminar Report Submitted to

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

*In Partial fulfilment of the requirements
for the award of degree of*

BACHELOR OF TECHNOLOGY

In

CIVIL ENGINEERING

By

ARYA AJI (SBW19CE012)

Guided by

Ms. ANUJA PRADEEP



**DEPARTMENT OF CIVIL ENGINEERING
SREE BUDDHA COLLEGE OF ENGINEERING
ELAVUMTHITTA, PATHANAMTHITTA 689625
DECEMBER 2022**

AUGMENTED AND VIIRTUAL REALITY IN CONSTRUCTION

SEMINAR REPORT

By

ARYA AJI (SBW19CE012)

Submitted to

APJ Abdul Kalam Technological University

in partial fulfilment of the requirements

for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

CIVIL ENGINEERING



**DEPARTMENT OF CIVIL ENGINEERING
SREE BUDDHA COLLEGE OF ENGINEERING
ELAVUMTHITTA**

**SREE BUDDHA COLLEGE OF ENGINEERING
ELAVUMTHITTA, PATHANAMTHITTA 689625
KERALA**

DEPARTMENT OF CIVIL ENGINEERING

CERTIFICATE



This is to certify that the seminar entitled “**AUGMENTED AND VIRTUAL REALITY IN CONSTRUCTION**” is a bonafide work done by **Arya Aji (SBW19CE012)**, under our guidance and supervision in partial fulfilment of the requirement for the award of the **Degree of Bachelor of Technology in Civil Engineering of the APJ Abdul Kalam Technological University** in the academic year 2022.

Guide	Seminar Coordinator	Head of the Department
Ms. Anuja Pradeep	Ms. Midhila V S	Ms. Midhila V S
Asst. Professor	Asst. Professor	Asst. Professor
Dept. of Civil Engg.	Dept. of Civil Engg.	Dept. of Civil Engg.

ACKNOWLEDGEMENT

I would also like to thank my guide **Ms. Anuja Pradeep**, Asst. Prof., Department of Civil Engineering for her valuable advice and wholehearted co-operation without which this seminar would not have seen the light of the day.

I express my sincere gratitude to **Ms. Midhila V S**, Asst. prof., Head of Department of Civil Engineering, Sree Buddha College of Engineering, Ayathil, Elavumthitta for encouragement and support without which this seminar could not have been accomplished.

I am grateful to the seminar coordinator **Ms. Midhila V S**, Assistant Professor and all my teachers of the Department of Civil Engineering & my friends for their valuable suggestions and encouragement.

Above all, I thank the almighty **GOD** without whose blessing I would ever have been able to complete this work successfully.

ARYA AJI

ABSTRACT

Augmented Reality (AR) and Virtual Reality (VR) are very effective tools for utilizing advancements in computation and technology in the construction sector, AR and VR have applications in the visualization of design, monitoring work progress, etc. This paper presents various advantages and challenges of AR and VR in the construction industry. The influence of VR in BIM methodology and its capacities were discussed. A systematic study of factors that drive and limit the adoption of these technologies.

A mixed research method combining qualitative and quantitative data collection was employed. VR & BIM capacities show how these methods influence the construction sectors and how ease to be work with them. Various challenges and advantages, which drive and limit the growth were identified and ranked. Eight focus groups with 54 experts and an internet-based questionnaire were conducted. 42 factors that drive and limit adoption were identified and ranked. The key challenges were grouped and classified to eliminate them as technology evolves. The major limitation of adoption is that AR and VR technologies are regarded as expensive and immature technologies that are considered to be as limitations. The prominent driving factors are that AR and VR enable improvements in project delivery and the provision of new and better services. Mitigation of limitations and encouraged adoption has to be promoted for increased adoption of AR and VR in the construction.

Keywords: Augmented Reality, Virtual Reality, Mixed Research Method

CONTENTS

Title	Page No.
ACKNOWLEDGEMENT	i
ABSTRACT	ii
LIST OF TABLES	vi
LIST OF FIGURES	vii
ABBREVIATIONS	viii
NOTATIONS	ix
1. INTRODUCTIONS	01
1.1 General	01
1.2 Augmented reality	01
1.3 Virtual reality	01
1.4 Mixed reality	03
2. USES OF AR & VR IN CIVIL ENGINEERING	04
2.1 General	04
2.2 Project planning	04
2.3 Automated measurements and error identifications	04
2.4 Project modifications	04
2.5 Safety training	05
2.6 On-site project modifications	05
2.7 Team collaborations	06

3. RESEARCH METHODOLOGY	06
3.1 General	06
3.2 Analysis	06
4. BIM METHODOLOGY WITH VR TECHNOLOGY	08
4.1 ND/BIM models	08
4.1.1 3D/BIM models	08
4.1.2 4D/BIM models	08
4.1.3 5D/BIM models	08
4.1.4 6D/BIM models	08
4.1.5 7D/BIM models	09
5. VR & BIM CAPACITIES	10
5.1 General	10
5.2 Walkthrough	10
5.2.1 Hologram projections	11
5.2.2 Enscape	11
5.2.3 Smart reality	13
5.2.4 Visidraft	13
5.2.5 CAVE	14
5.3 Consulting data	15
5.3.1 Autodesk 360	16
5.3.2 Viewer revit plugin	16
6. ADVANTAGES AND LIMITATIONS FOR ADOPTING AR & VR COMPARED WITH BIM METHODOLOGY	17
6.1 General	17

6.2 Advantages	17
6.3 Limitations	17
6.3.1 Immature technologies	18
6.3.2 Non-technical issues	18
6.3.3 Special requirement for implementations	19
6.3.4 Sectoral and client dynamics	19
7. DRIVERS FOR ADOPTING AR & VR IN CONSTRUCTION INDUSTRY	20
7.1 General	20
7.2 Improving performances in projects	20
7.3 Improving companies image	20
7.4 Improving companies overall performances	20
7.5 Bolstering research and development	21
8. CONCLUSIONS	22
REFERENCE	23

LIST OF FIGURES

No.	Title	Page. No.
Fig. 1.1	VR view of a client at different perspective	2
Fig. 1.2	VR HMD	2
Fig. 1.3	Mixed Reality Spectrum	3
Fig. 3.1	Diagram of the mixed research methodology used in this study	6
Fig. 4.1	7D/VR/BIM applications	9
Fig. 5.1	Virtual reality tour inside and outside a BIM model	10
Fig. 5.2	Touchable holograms	11
Fig. 5.3	Walkthrough capacity using Enscape Revit plugin	12
Fig. 5.4	Smart Reality application	13
Fig. 5.5	Visualizing, in mobile device, a 3D model using Visidraft app	14
Fig. 5.6	An igloo BIM CAVE	14
Fig. 5.7	Using Autodesk 360 in a collaborative project	16

ABBREVIATIONS

AR	Augmented Reality
VR	Virtual Reality
HMD	Head-Mounted Display
MR	Mixed Reality
AEC	Architecture Engineering and Construction
MTC	Manufacturing Technology Centre
BIM	Building Information Modelling
FGD	Focus Group Discussion
HVAC	Heating Ventilation and Air Conditioning
CAVE	Cave Automatic Virtual Environment
PCA	Principal Component Analysis
R&D	Research and Development

NOTATIONS

\$	Dollar
i.e.,	That is
%	Percentage

1.INTRODUCTION

1.1 GENERAL

With the advent of Augmented reality (AR) and virtual reality (VR) technologies, there is a huge turn-around in the way humans interact with visual information. Every industry is seeing huge adoption of these technologies and it is regarded as one of the top 10 Gartner strategic technology trends for 2019. By 2025, AR and VR are projected to become an \$80 billion market. Yet Architecture, Engineering, and Construction sectors are seeing a much slower pace of growth in these emerging technologies. Digitization in the construction industry is rated as the lowest of 22 industries.

The lower level of adoption of these trends in the construction industry has to be studied systematically. The specific objectives are:

- 1) To define use cases of AR and VR in construction.
- 2) To provide Qualitative and Quantitative analysis of usage of these technologies.
- 3) To identify advantages of AR, VR adoption and factors that limit it.

1.2 AUGMENTED REALITY

Augmented Reality (AR) is an interactive experience that combines the real world and computer-generated content. AR can be defined as a system that in cooperates three basics features: a combination of real and virtual worlds, real-time interaction and accurate 3D registration of virtual and real objects. The overlaid sensory information can be constructive (i.e., additive to the natural environment), or destructive (i.e., masking to the natural environment). This experience is seamlessly interwoven with the physical world such that it is recognized as an immersive aspect of the real environment. In this way, augmented reality one's ongoing perception of real-world environment, virtual reality replaces thar user's real-world environment.

1.3 VIRTUAL REALITY

Virtual Reality (VR) is a simulated experience that employs 3D near-eye to give the user an immersive feel of the virtual world. Applications of virtual reality include entertainment, education, and business. Other distinct types of VR-style technology include augmented reality, mixed reality, sometimes referred to as extended reality although definitions are currently changing due to the commencement of time of the industry.

Currently, standard virtual reality systems use either virtual reality headsets or multi-projected to generate realistic images, sounds and other sensations that stimulate a user's physical presence in a virtual environment.



Fig. 1.1. VR view of a client at different perspective

A person using VR equipment is able to look around the artificial world, move around in it, and interact with virtual items. The effect is commonly created by VR headsets consisting of a Head-Mounted Display (HMD) with a small screen in front of the eyes but can also be created through specially designed rooms with multiple large screens. VR typically incorporates auditory and video feedback but may also allow other types of sensory and force through haptic technology.



Fig. 1.2. VR HMD

1.4 MIXED REALITY

VR enables us to create an entirely computer-generated environment. It creates a perception of a 3D artificial environment. Virtual experience is an essential part VR and is provided using HMD.

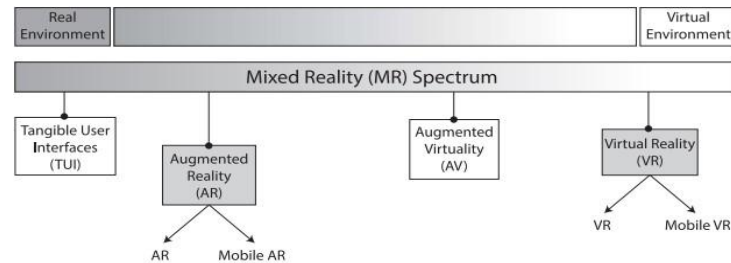


Fig. 1.3 Mixed Reality Spectrum

However, AR provides an overlay of virtual information over real environments which can be viewed through handheld devices or smartphones. MR anchors virtual objects into the real environment.

2. USES OF AR & VR IN CIVIL ENGINEERING

2.1 GENERAL

Augmented Reality has widespread applications in architecture and design and adds enormous value to the architectural projects by effectively portraying what exactly an architect is trying to convey. A major AR help to architects is the ability to showcase architectural models in a totally new way. Let us now evaluate how architects use AR to create superior designs and persuade their clients better.

2.2 PROJECT PLANNING

Combined with 3D demonstration software and Building Information Modelling (BIM), Construction companies can create detailed, interactive models of building projects and present them to customers towards the beginning of a project. This permits customers to get a realistic perspective on the outcome and suggest any ideal improvements before the construction starts.

2.3 AUTOMATED MEASUREMENTS AND ERROR IDENTIFICATION

Augmented reality gear can measure a space's actual properties, including its dimensions. Construction companies can incorporate this information into models, permitting them to produce considerably more precise constructions and have a more comprehensive view of how the project will look like. Real-time error identifications can be done and can be corrected accordingly. This permits workers to find any inconsistencies in the structures and quickly adjust them to prevent higher costs and delays down the line.

2.4 PROJECT MODIFICATIONS

One significant advancement in AR and VR is the ability for users to make changes to building models directly on the site. This application is significantly helpful, as it considers specialist to investigate any mistakes in a virtual view prior to applying changes to the actual structure. Since this computerized information is constantly refreshed, improves the workflow, and prevents any wastage of materials.

2.5 SAFETY TRAINING

Safety is a major concern in the construction industry. With the assistance of an AR headset, laborers can be shown virtual drills, guidance, and security situations. This style of direct learning not just improves their awareness of safety and provides intuitive training but lowers training costs and downtime.

2.6 ON-SITE PROJECT MODIFICATIONS

The ability to combine all digital information with one's physical view is a vital part of AR and VR. Detailed information regarding the location of pipes, walls, outlets, switches, and ventilation can be assessed directly on the site in layers that can be easily tagged to show or hide. Information retrieval allows field workers to efficiently monitor a project against its building plan and ensure successful project completion.

2.7 TEAM COLLABORATION

Field workers can share real-time data and project progress to remote team members. Remotely manage and analyse the project progress and can make desired changes without actual site visit.

3. RESEARCH METHODOLOGY

3.1 GENERAL

A mixed research method, comprising of qualitative and quantitative data collection is utilized for this study.

3.2 ANALYSIS

It is found to be useful in identifying underlying factors in complex system by the support of quantitative data collection with qualitative research activities. Qualitative analysis is conducted by extrapolatory workshops with experts in the field, with a series of Focus Group Discussion (FGD), FGD were used to collect the opinion of experts in the field regarding factors that limit and drive the adoption of AR and VR. Factors that drive and limit the penetration of AR and VR were defined. The main activities conducted in this part of the study were four extrapolatory workshop, in which two FGDs, were carried out in each workshop level. In total, eight FGDs were held, with duration between 30 and 45 minutes, 54 experts from 36 organizations, companies and academic institutions based in the UK participated in workshops.

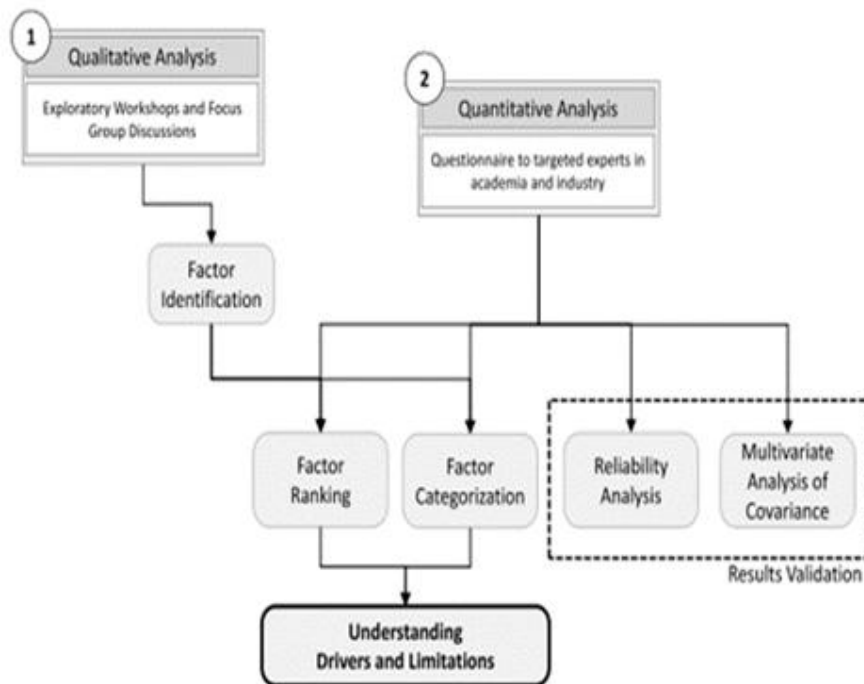


Fig. 3.1. Diagram of the mixed research methodology used in this study

From qualitative analysis, 21 driving factors for the adoption of AR and VR in the construction industry were identified. The questionnaire was developed to quantify the

factors on the 'Linkert's scale' which ranks the factors from 1 (lowest importance) to 5 (highest importance). The questionnaire was pilot tested by experts to ensure the clarity of the question and the structure and logic of the questionnaire. Reliability analysis of the result is done to validate the internal consistency.

4. BIM METHODOLOGY WITH VR TECHNOLOGY

4.1 ND/ BIM MODELS

BIM methodology can generate and maintain information produced during the whole lifecycle of a building project, from design to maintenance, in a centralized model, the BIM model. As so, the BIM data-rich model support the application of the rich-data in various field. The model allows view and data, appropriate to various users need. Distinct type of data can be extracted from the model, analysed, and applied to generate new information. Due to the consistency of design data with quality data and construction modelling process with value control, the potential of BIM implementation in improving management lies 60 State of the Art Virtual Reality and Augmented Reality Knowhow in its ability to present multi-dimensional data concerning design data. The use of BIM data in the development of several tasks is considered for the generation of ND/BIM applications.

4.1.1 3D/BIM models

3D/BIM models concern modelling process of all disciplines (architecture, structures, and MEP) and as it's based on parametric object, the relationships (spatial geometry) and the properties (physical and mechanical).

4.1.2 4D/BIM models

4D/BIM models refers to the construction planning supporting the visual simulation of the building construction and its control. The 4D application links 3D elements, organized in sets, to the correspondent time schedule, allowing the comparison between the real work and the planned sequence of the process (supports eventual changes and allows a permanent actualization of schedules, estimates, and logistic site management).

4.1.3 5D/BIM models

5D/BIM model is associated with cost estimation (control of the financing plan based on take-off material quantities in each constructive step).

4.1.4 6D/BIM models

6D/BIM model supports the performance of energy and sustainability studies (energy consumption estimate along the project and subsequent monitoring, measurement and verification of consumption during the occupation of the building).

4.1.5 7D/BIM models

7D/BIM model is generated to support the maintenance activity and distinct management facilities needed in the building during its occupancy. Combining BIM with VR is predictable to improve efficient collaboration, as the data can be visualized, consulted, and even changed while the team and client are inside an apparently construct build. In a real place, the equipment that cannot be observed because they are hidden over a false ceiling, such as HVAC devices or net systems (electrical, water, or gas supply or elements of acoustic protection), can be easily visualized using BIM + VR tools. So the management and maintenance facilities can be facilitated in a 7D/VR + BIM/model context. VR will enable the engineer to carry out a physical survey of the property or the building site. The VR technology applied to BIM tools has been enhancing a high level of integration, but it is still necessary to do more in order to obtain an even more real platform, in a retrieving data perspective.



Fig. 4.1. 7D/VR/BIM applications.

As so, the importance of adding VR benefits should be a constant research point. The maintenance aspect needs the user to free consult all components of the real place through the correspondent BIM model. This information should be able to be consulted from the model, and the access to this information could be easily provided. Any BIM tool allows the consult of BIM data and so it support management and maintenance within a collaborative team.

5. VR & BIM CAPACITIES

5.1 GENERAL

VR technology leads to better communication for key players across the building sector, based on greater design visualization, contributing to a better understanding of the project. Interacting with the BIM model, to visualize the geometry of the elements and to consult the parametric data, is an improvement in the development of a collaborative project. The acceptance of VR use in construction field has been growing. VR enables the user to fully immerse himself in a 1:1-scale, 3D/BIM model which can be manipulated, providing an immersive accurate sense of presence in a space that is yet to be built. AEC management professionals recognize that VR applications are making it easier for clients to visualize designs earlier, saving material costs off budgets and reducing the number of workers needed for projects. For instance, a VR tour can be applied over a BIM model to check, from a facility management perspective, the maintenance schedule (7D/BIM model), or from a project team view, the constructability review (4D/BIM model), supporting decisions making. VR technology can improve BIM methodology, as it allows interaction with 3D/BIM models in two essential ways:

5.2 WALKTHROUGH

Virtual reality can improve the development and analyses of any detail of the project, along distinct design steps, as it easily involves all team, and a justifiable decision can be worked out concerning alternative solutions or problems resolution.



Fig. 5.1 Virtual reality tour inside and outside a BIM model

VR offers more realistic possibilities as the final model can be rendered in 3D and the team member can experience and explore the space as a real space. VR technology allow the user to experience artificial world, by walking through outside and inside and see everything around and reach out touching object in real time. To create an immersive VR environment computer graphics technology and a selective equipment with advanced viewing and interaction capabilities are needed. The 3D virtual world used to support the distinct aspects of the construction activity is permanently increasing nowadays.

The current perspective, reported in research works and in commercial web sites, is the use of VR applied over the BIM model, as a positive and an innovative contribute, to improve the construction industry efficiency. There is almost 50+ VR software and hardware tools available right now that can redefine the way BIM is implemented and interpreted. Currently, there are various applications of virtual reality tools that can be applied in the construction activity as a support to engineers' and architects' work which includes:

5.2.1 Hologram Projections

The use of hologram projections in the construction activity, enables the end user to experience different textures or choose from a variety of in-built texture formats the required aspect for the project. The Revit model can be manipulated using hologram projections supporting a building collaborative design.



Fig. 5.2 Touchable holograms

5.2.2 Enscape

With the Enscape Revit plugin, the user is able to quickly explore different design options and present projects to clients and so it has become a standard application in projects worldwide. A walkthrough using Enscape usually runs alongside Revit or

SketchUp, reflecting all the perspectives, geometric details, and eventual adjustments to the project.



(a)



(b)

Fig, 5.3 Walkthrough capacity using Enscape Revit plugin

Using Enscape for real-time visualization and rendering enables the project team to accurately communicate with engineering design, clients, and stakeholders, improving the collaborative design. The ability to walkthrough inside BIM models and to regenerate design changes without leaving Revit is of a great importance and time saver. So Enscape add a significant value to communication in a team, as a real-time rendered walkthrough, is able to be generated straight from Revit models. In this immersive and highly realistic way, consultants or designers better understand the project, as they can judge not just the technical aspect but also the realistic visual impact.

5.2.3 Smart Reality

Smart Reality app brings a new realism of tools to interact with clients, engineers, contractors, and other professionals in the building industry. The user is instantly immersed in a true 3D environment that gives a sense of scale, depth, and spatial awareness adding a more realistic perspective of the model and its environment. The VR mobile app was developed for builders and combines 2D plans and 3D project models for an enhanced, mobile visualization of construction projects. The VR capacity of this app is beginning to introduce a new level of certainty for the design and construction process as well, helping stakeholders to get a fuller experience of the BIM model, offering an unobstructed, real-world perspective of projects for planning, design, and construction management. It is designed for the AEC sector and works with BIM tools like Autodesk's Revit. As the design process is slow, Smart Reality can be used to help the owner or architect to get a better feel of spatial recognition of the intended product in advance of construction beginning. The viewing experience can further be enhanced by using VR headsets like Oculus Rift and Epson's Move Rio BT-200 smart glasses.



Fig. 5.4 Smart Reality application

5.2.4 Visidraft

The Visidraft app builds a 3D world around the user. Visidraft instantly converts 2D elevations as well plans into corresponding 3D projections (when viewed through a mobile camera). When the user places virtual objects into the environment, the app automatically calculates the relative distances of all items present in that room allowing

the user to walk around the room physically with the device and visualize and re-arrange the items conveniently.



Fig. 5.5 Visualizing, in mobile device, a 3D model using Visidraft app

5.2.5 CAVE

CAVE (cave automatic virtual environment) platforms have been developed for immersive VR experience as they track user's head and control stick usually with 6 degrees of freedom, to navigate inside the virtual environment and interact with the contents of a 3D model including a 3D BIM model. A CAVE normally takes the form of a cubelike space in which images are displayed by a series of projectors.



Fig. 5.6 An igloo BIM CAVE

Due to its immersive experience and intuitive manipulation capability, it quickly gained popularity in both research and construction industry community. As the team member better communicate between them and the client better understand spatial relationships

and scale of the project. VR can induce a quit realist analyses as the physical construction seems to be present. VR plays an important role at all stages of the design to-construction process. Igloo BIM cave has space constraints; there are limitations on where an environment can be placed and how easy it is to bring people to it.

5.3 CONSULTING DATA

Once the model is inside the VR environment, things like materials, lighting, furniture, and other small details that make the VR experience feel real, are added. This is the concept of walkthrough, but BIM contains data and it is convenient to explore BIM data while walking inside a virtual model. VR can play an important role at all stages of the design-to-construction process, from evaluating design options and showcasing proposals, to designing out errors and ironing out construction and serviceability issues before breaking ground on site. Fully interactive VR software has extremely high-performance demands, during a tour, but additional technology capacities concerning model-data retrieving is required when bringing BIM data into a VR environment.

Collaboration for BIM tools is designed to help project teams overcome the barriers of corporate firewalls and physical location by enabling centralized access to BIM models. Team members in all disciplines from multiple companies or sites worldwide can access and work in models hosted on cloud servers. The model is centrally hosted and access to project data is permissions based, and it can be accessed by project teams anywhere. So, when collaborative projects are discussed between firms, they consult BIM data in models hosted on clouds. In addition, the capacity of consulting BIM data can be improved with VR devises applied directly over the BIM model. Combining VR devices applied over the BIM model and the real inspection in the place, it offers a new perspective for facility management professionals. VR has the capacity to improve how managers and construction workers access virtual information of the established project. All data-rich virtual information is available from the BIM model, and can be retrieved, consulted, and analysed during the real inspection to the site. Construction issues can be addressed in real-time and internal assets can be illustrated to facilitate installation and layout, saving time, and resources. Several plugins or BIM viewer allows consulting data from the 3D/BIM model and now there is a new class of mobile apps available that let project teams experience building plans in 3D.

6. ADVANTAGES AND LIMITATIONS FOR ADOPTING AR & VR COMPARED WITH BIM METHODOLOGY

6.1 GENERAL

Since AR & VR are considered as the fast-growing technology in the construction industry, its advantages and limitations for the adoption were also mentioned.

6.2 ADVANTAGES

Visualizing BIM models in virtual reality enhances all the possibilities your model has and optimizes its perception including important aspects such as depth, spatial sensation, amplitude perception and proportion between the different elements, impossible to achieve by working with a BIM model on a computer screen.

The advantages are:

- (i) Exponentially improves the understanding of the model and its characteristics by all the agents involved in the project.
- (ii) Interacts with materials and objects in an immersive way–
Improve detection of failures and architectural or structural elements, checks BIM models to discover structural inefficiencies, clash detection, modelling, or design errors, in addition to detecting opportunities in the project, in the monitoring of work and in the maintenance of the building.
- (iii) Obtain great possibilities of personalization.
- (iv) Acquire a competitive difference showing your projects.
- (v) Possibility of process simulation (trainings or virtual trainings).
- (vi) Make models more attractive and understandable by improving your customers' perception and reducing marketing costs: the customer will be able to “walk through” your model it is built anywhere in the world.
- (vii) Manage your model in an interactive way.
- (viii) Increase productivity and quality of projects.
- (ix) Make changes possible and receives feedback in real time.

6.3. LIMITATIONS

Using data reduction tool PCA (Principal Component Analysis), four limiting factors

were extracted. The categories are ordered based on their percentage of variance and not on the importance of their individual factors.

6.3.1 Immature Technologies

AR and VR are still immature technologies that cannot be fully used in practice yet. Battery limitations, narrow fields of view, low tracking accuracy, low resolution, and uncomfortable HMDs, etc often discourage the use of these in the AEC sector. Very complex 3D models used in the construction industry cannot be handled by current AR/VR devices. Since research and development in AR and VR aim mainly in its development in the entertainment sector, it lacks the precision, accuracy, consistency, and efficacy needed for the engineering sector.

Building Information Modelling (BIM) and AR/VR find collaborative implementation in the field of construction. The rate at which BIM is being adopted varies globally. With India standing amongst the lowest (with just 10-18%)(Jagadeesh n.d.), adoption of AR/VR will be very difficult. The use of the smart and computable 3D model of the project developed by BIM software increases the applications of AR and VR. Lack of adoption of BIM inhibits AR and VR adoption as well. Regarding labour dynamics, the construction industry is not yet a mature field, unlike the entertainment industry. It is very difficult for construction firms to attract AR and VR experts, who usually prefer the entertainment and gaming industries.

There are not enough graduates with the required skills, and only large firms manage to have teams dedicated to AR and VR development. In addition, the workforce in the construction industry needs to be upskilled, which might represent a huge challenge, because in general the construction industry does not have a trained workforce, and upskilling has not been widely employed.

6.3.2 Non-Technical Issues

The second category in importance includes factors related to non-technical issues that limit adopt. Construction firms do not have expertise and knowledge in AR and VR hardware, software, and its dynamics. It is difficult for companies to compare various devices and technologies available in the market. Being perceived as technologies for the entertainment sector, many believe that these technologies don't have much potential in the AEC sector. However, this notion considers the huge advantages augmentation can bring to this sector. As with every other digital technology, issues with the aversion to change, job insecurity, data security, and data ownership exist. And

such issues are more pronounced in the construction industry because of low digitization and lack of a trained workforce. A sudden implementation may not be possible.

6.3.3 Special Requirement for Implementation

Expensive hardware, software, management system, trained workforce, etc are essential for the implementation. VR requires HMDs (Head Mount Displays), movement tracking sensors, special controllers, powerful computing devices with high-end graphics performance. These equipment's requires huge investment for wide adoption. This will cause a huge burden for small/medium companies. Also, the return of investment will not be instantaneous.

An information exchange system, data management, and connected environment are essential to have a seamless experience of these technologies. Lack of capabilities is amplified by poor data management and data exchange practice in the construction industry. Multiuser capabilities in AR/VR are still under development. Since construction firms are not working on an individual basis, every task requires group effort and collaboration. Multiuser AR is limited by the bandwidth of the wireless connection, especially in developing countries with slower data speeds is a matter of concern. The lack of multiuser capabilities is the reason that AR and VR are regarded as technologies that can improve communication but also increase isolation and inhibit collaboration.

6.3.4 Sectoral and Client Dynamics

The AEC sector is highly fragmented, adoption of emerging digital technologies is a difficult task. Construction projects are usually carried out by a group of small and large companies. From procurement of raw materials to completion of work multiple companies may be engaged and multiple subcontracts may be given. The successful adoption of digital technologies like AR and VR is only possible in industries with lower fragmentation and tight control. Lack of client interest is another limiting factor. Increased client requirements can promote the use of new-age tech in the construction sector. But considering the conservative character of the construction sector, adoption of these technologies is not considered a priority.

7. DRIVERS FOR ADOPTING AR & VR IN CONSTRUCTION INDUSTRY

7.1 GENERAL

Four categories of driving factors were extracted from the study.

7.2 IMPROVING PERFORMANCES IN PROJECTS

Improved communication and collaboration, improved project understanding, improved productivity, and reduced spending on projects, etc can be achieved by effective utilization of AR and VR. AR and VR usage shows a steady increase in the AEC sector in the USA. A 26 percent increase in usage is observed between 2017 and 2018(Noghabaei et al. 2020). This increase in adoption is driven by improved product delivery and reduced cost.

7.3 IMPROVING COMPANIES IMAGE

Adoption of newer technologies can improve the image of the company. It can brand itself as an innovating company and that gives a differentiating advantage in the competitive market. Construction companies are rebranding themselves as smart engineering solutions providers. With digital giants like IBM and Oracle venturing into the construction sector, their expertise in digital technologies has great potential in the AEC sector.

7.4 IMPROVING COMPANIES OVERALL PERFORMANCE

This category incorporates the variables addressing the longing of organizations to make progress toward the achievement of organizational goals. The adoption of AR and VR advancements is viewed as an opportunity to offer new and better types of assistance and to grow to other business sectors. For instance, organizations in areas going from business consultancies to retail and magnificence items are contributing essentially to AR and VR advancements. It likewise is viewed as a way of working on the association's work culture and to build overall productivity.

The reception of any arising computerized innovation can contribute to further developing work culture in associations. The reception of AR and VR additionally is viewed as a way of decreasing hazards, which is vital for the construction sector, which is viewed as a high-hazard area.

7.5 BOLSTERING RESEARCH AND DEVELOPMENT

Bolstering research and development (R&D) is the last category of factors that drive adoption, which incorporates factors Fostering the research interest of the workers and Increasing R&D interest in the development area. Increasing R&D investment in the construction sector drives the adoption of AR and VR. Globally, the R&D investment in the AEC sector has doubled in the last decade. The degrees of R&D interest in the development industry should proceed and speed up to catch up with that of the leading sectors.

8. CONCLUSIONS

This paper presented an idea about Augmented reality, Virtual reality and mixed reality. The uses of these technology in the construction industries were also extracted. Mixed research investigation of the elements that drive the adoption of AR and VR technologies and the challenges faced. Important factors were ranked and quantified. Four categories of limiting and driving factors have been defined. BIM methodology can influence VR technology in various ways ranging from 3D/BIM models to 7D/BIM models. Also, VR capacities can be achieved through various applications, websites as well specially designed equipment's.

This review shows that AR and VR can be utilized in different ways throughout the entire life cycle of a built asset. Still the overall adoption in the industry remains low. Regardless of the low degrees of adoption, there is a sign that construction organizations have an interest in putting resources into AR and VR technologies. It is observed from the study, AR and VR are not ready to be fully adopted in the construction industry and that research and development gaps remain. Likewise, it presents a powerful fundamental review that sets the course for the examination important to empower the fruitful reception of AR and VR in the construction industry.

REFERENCE

1. **Alcínia Zita Sampaio** (2016) “Enhancing BIM Methodology with VR Technology” *Additional information is available at the end of the chapter* <http://dx.doi.org/10.5772/intechopen.74070> © The Author(s). Licensee Intech.
2. **Davila Delgado, Juan Manuel, Lukman Oyedele, Thomas Beach, and Peter Demian.** (2020). “Augmented and Virtual Reality in Construction: Drivers and Limitations for Industry Adoption.” *Journal of Construction Engineering and Management* 146(7):04020079. doi: 10.1061/(asce)co.1943-7862.0001844.
3. **Davila Delgado, Juan Manuel, Lukman Oyedele, Peter Demian, and Thomas Beach.** (2020). “A Research Agenda for Augmented and Virtual Reality in Architecture, Engineering and Construction.” *Advanced Engineering Informatics*.
4. **Dr. K. Chandramouli** (2021) “A Study on E-Highway-Future of Road Transportation” *International Journal of Engineering and Advanced Technology* (IJEAT); ISSN: 2249 – 8958, Volume-8, Issue-2S2, (2; SCOPUS INDEXED
5. **Eadie R, Browne M, Odeyinka H, McKeown, McNiff S.**(2013) “BIM implementation throughout the UK construction project lifecycle” *An analysis. Automation in Construction*. 2013;36:145-151. DOI: 10.1016/j.autcon.2013.09.001
6. **James Garbett¹, Thomas Hartley^{1^} and David Heesom^{2*}** ¹ (2019) “A Multi-User Collaborative BIM-AR System to Support Design and Construction” School of Mathematics and Computer Science, Faculty of Science and Engineering.
7. **K.Purushottam¹, Dr. K. Chandramouli², J.Sree Naga Chaitanya³, B. Gowreswari⁴** (2020) “A Review on Virtual Reality and Augmented Reality in Architecture, Engineering and Construction Industry” *journal on Construction Engineering and Management*
8. **Noghabaei, Mojtaba, Arsalan Heydarian, Vahid Balali, and Kevin Han** (2020). “Trend Analysis on Adoption of Virtual and Augmented Reality in the Architecture”, *Engineering, and Construction Industry*. Data 5(1). doi: 10.3390/data5010026
9. **Poussard B, Loup G, Christmann O, Eynard R, Pallot M, Richir S.** (2014) “Investigating the Main Characteristics of 3D Real Time Tele-Immersive Environments through the Example of a Computer Augmented Golf Platform”, Laval Virtual VRIC ‘14, April 9-11, Laval, France. Article No. 31. DOI: 10.1145/2617841.2620720

- 10. Sepehr Alizadehsalehia, Ahmad Hadavib, Joseph Chuenhuei Huang**
Corresponding author. (2021) “From BIM to extended reality in AEC industry.”